EVERY PROPER MODIFICATION THAT IS AN ISOMORPHISM OVER A DENSE OPEN IS DOMINATED BY AN ADMISSIBLE BLOW-UP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9003-9016 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9018-9022, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Two-move composition with the engine cited by name; standard reduction, accepted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma isomorphism-after-blowing-up (the blow-up that trivializes a proper isomorphism-over-open situation); functoriality of strict transforms.

Source reference. Original lemma 0GUW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8937–8945. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Let $U \subset B$ be an open subspace. Assume*

(1) *B is quasi-compact and quasi-separated,*
(2) *U is quasi-compact,*
(3) *$f : X \to B$ is proper,*
(4) *$f^{-1}(U) \to U$ us an isomorphism.*

Then there exists a U-admissible blowup $B' \to B$ which dominates X, i.e., such that there exists a factorization $B' \to X \to B$ of the blowup morphism.

Proof. By Lemma 0GUW we may find a U-admissible blowup $B' \to B$ such that the strict transform X' maps isomorphically to B'. Then we can use $B' = X' \to X$ as the factorization. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.